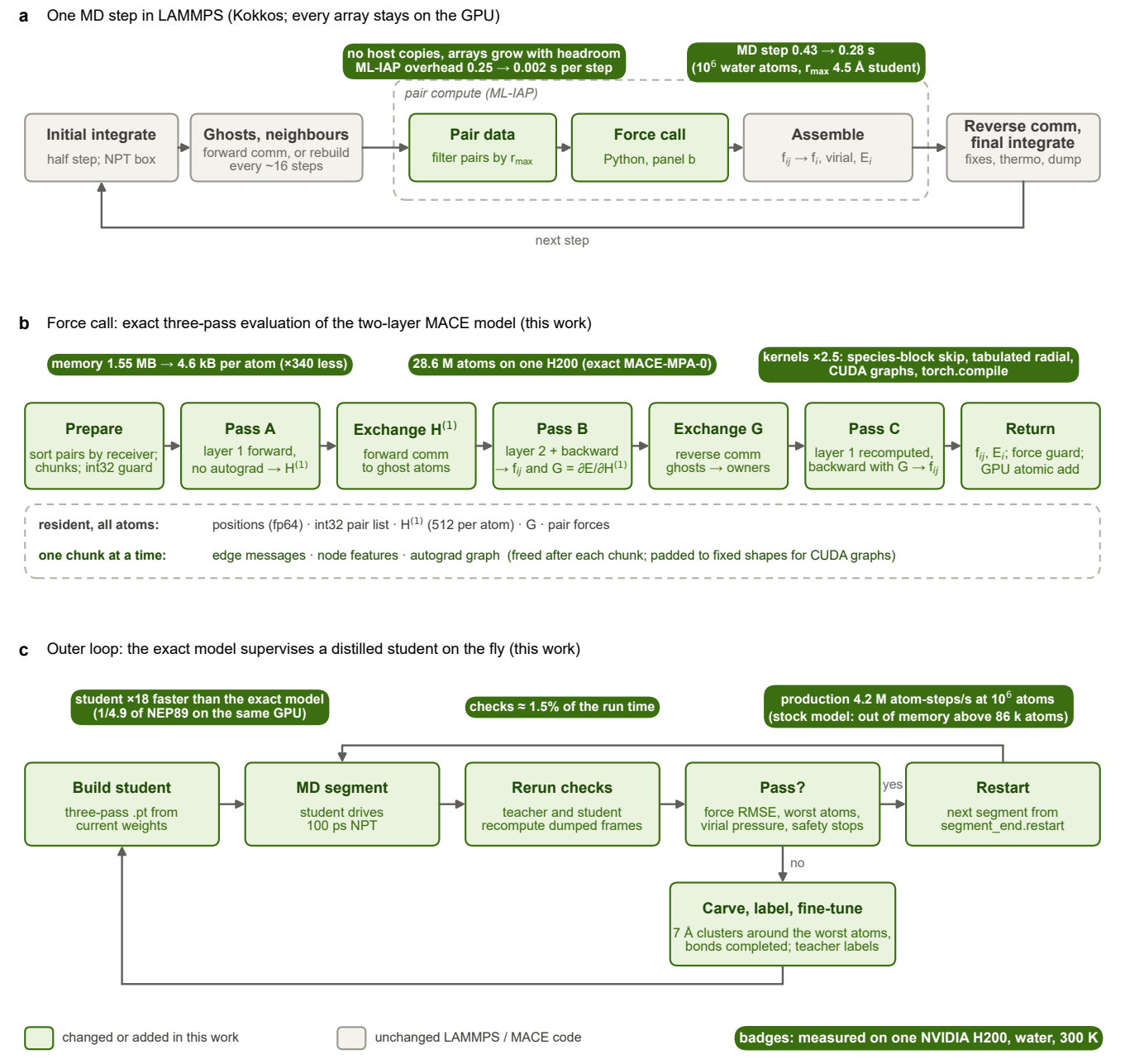

a One MD step in LAMMPS (Kokkos; every array stays on the GPU)
no host copies, arrays grow with headroom
ML-IAP overhead 0.25 → 0.002 s per step
MD step 0.43 → 0.28 s
(10^6 water atoms, r_max 4.5 Å student)
pair compute (ML-IAP)
Initial integrate
half step; NPT box
Ghosts, neighbours
forward comm, or rebuild every ~16 steps
Pair data
filter pairs by r_max
Force call
Python, panel b
Assemble
f_ij → f_i, virial, E_i
Reverse comm, final integrate
fixes, thermo, dump
next step
b Force call: exact three-pass evaluation of the two-layer MACE model (this work)
memory 1.55 MB → 4.6 kB per atom (×340 less)
28.6 M atoms on one H200 (exact MACE-MPA-0)
kernels ×2.5: species-block skip, tabulated radial, CUDA graphs, torch.compile
Prepare
sort pairs by receiver; chunks; int32 guard
Pass A
layer 1 forward, no autograd → H(1)
Exchange H(1)
forward comm to ghost atoms
Pass B
layer 2 + backward → f_ij and G = ∂E/∂H(1)
Exchange G
reverse comm ghosts → owners
Pass C
layer 1 recomputed, backward with G → f_ij
Return
f_ij, E_i; force guard; GPU atomic add
resident, all atoms: positions (fp64) · int32 pair list · H(1) (512 per atom) · G · pair forces
one chunk at a time: edge messages · node features · autograd graph (freed after each chunk; padded to fixed shapes for CUDA graphs)
c Outer loop: the exact model supervises a distilled student on the fly (this work)
student ×18 faster than the exact model
(1/4.9 of NEP89 on the same GPU)
checks ≈ 1.5% of the run time
production 4.2 M atom-steps/s at 10^6 atoms
(stock model: out of memory above 86 k atoms)
Build student
three-pass .pt from current weights
MD segment
student drives 100 ps NPT
Rerun checks
teacher and student recompute dumped frames
Pass?
force RMSE, worst atoms, virial pressure, safety stops
yes
Restart
next segment from segment_end.restart
no
Carve, label, fine-tune
7 Å clusters around the worst atoms, bonds completed; teacher labels
changed or added in this work
unchanged LAMMPS / MACE code
badges: measured on one NVIDIA H200, water, 300 K